

Ella Boland

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Education

Stevens Institute of Technology

- Major: Mechanical Engineering
- Expected Graduation Date: May, 2028
- Cumulative GPA: 3.702

Work Experience

Stevens Institute of Technology

August 2024 - Current

Researcher

- Partnered with faculty to provide research on a new video game
- Coded the background and nonplayer characters of the game.

Stat Caller

- Gathered stats for men's volleyball games
- Assisted athletic department during matches

McHenry Community High School

August 2023 - May 2024

Teaching Assistant - Engineering

- Created robotic projects for the engineering students
- Classroom organization and assistance to Engineering professor

Technical Proficiencies and Skills

- CAD Modeling (OnShape, Solidworks, Fusion 360)
- Experience with coding (C++, MATLAB)
- Proficient with 3D printing

Past Projects

College Freshmen Boat Design Project

- The project goal was to design a small plastic boat with a group that was programmed to do three laps around a small pool in under a certain set time.
- Designed the boat's hull in order for it to hold all materials and remain able to float alone.
- Helped with the coding and setting up the circuitry for the boat.
- The end result was the successful completion of the three laps in time allowed.

College Sophomore Playground Project

- Worked with a team to fully 3D model different aspects of a children's playground and also calculate the force applied to different parts in order to ensure safety.
- Used resources like SolidWorks and Excel in order to model and organize data.

High School Engineering Project

- Chose the problem of comfortability of prosthetics and worked towards finding a realistic solution.
- The end solution and project was a prosthetic sleeve that used a hexagon cushioning design to help with shock absorption for a leg prosthetic while walking.
- Used Fusion 360 to model a mold for making a silicon sleeve and the actual sleeve design.

Volunteerism

Volunteer for Football and Basketball Concessions-NHS and Student Council

Aug 2022 – May 2024

- Concession cashier, food runner, and food preparer.

Volunteer at Youth Volleyball Camp for Middle School

May 2022 – May 2024

- Coached and taught middle schoolers 6th-8th grade basic volleyball skills.

Honors/Awards

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| • Dean's List for Stevens Institute of Technology (Fall 2024, Spring 2025, Fall 2025) | • Illinois State Scholar |
| • Stevens Pinnacle Scholars Program | • Volleyball Fox Valley All-Conference Team (Twice) |
| • MAC Second Team All-Conference | • High School Volleyball MVP |
| • Academic All-District | • Spanish Honor Society |
| • National Honor Society | • Distinguished Warrior |

Extra Curriculars

- Stevens Women's Volleyball Team
- Stevens Society of Women Engineers
- Sigma Delta Tau Sorority
- High School Volleyball Team (4 Years)
- Club Volleyball Team (6 Years)
- National Honor Society
- Spanish Honor Society
- High School Student Council (Class Officer 3 Years, Class President 1 Year)
- High School Spirit Rally Committee
- High School Renaissance Team (Group of students working with the principal to better the school)
- High School Connections Crew (Worked with Sophomore class to make them feel more involved)
- High School Principal's Council (Worked with Principal and small group to discuss school concerns)